

Functions & Graphs



Function transformations

- 1 Let $f(x) = x^2$ and $g(x) = -3f(x + 2) - 4 = -3(x + 2)^2 - 4$. Find $g(0)$, and describe the transformation from f to g in words.
 - 2 Using the same $g(x) = -3(x + 2)^2 - 4$, find $g(-2)$, and explain why this is the vertex of g 's graph.
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Even and odd functions

- 3 Classify $f(x) = x^4 + 2x^2 - 5$ as even, odd, or neither.
 - 4 Classify $h(x) = 2x^5 - 3x$ as even, odd, or neither.
 - 5 Explain why $k(x) = x^2 + x$ is neither even nor odd.
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Domain restrictions

- 6 Find the domain of $f(x) = \sqrt{2x + 6}$.
 - 7 Find the domain of $g(x) = \frac{1}{x^2 - 25}$.
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Piecewise functions

- 8 Let $p(x) = 2x + 3$ for $x < 1$, and $p(x) = x^2 - 1$ for $x \geq 1$. Find $p(-2)$ and $p(3)$.
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Multiple choice — easy

- 9 Classify $f(x) = 2x^3 + 5$ as even, odd, or neither.
 - a. even
 - b. both even and odd
 - c. neither
 - d. odd

- 10 Classify $f(x) = 3x^3 + x$ as even, odd, or neither.
- a. both even and odd
 - b. even
 - c. neither
 - d. odd
- 11 Classify $f(x) = -3x^5 + 6x$ as even, odd, or neither.
- a. both even and odd
 - b. even
 - c. neither
 - d. odd
- 12 Classify $f(x) = 3x^2 + 2$ as even, odd, or neither.
- a. both even and odd
 - b. neither
 - c. even
 - d. odd
- 13 Classify $f(x) = -x^2 + 6$ as even, odd, or neither.
- a. both even and odd
 - b. neither
 - c. odd
 - d. even

- 14 Classify $f(x) = 2x^5 + 2x$ as even, odd, or neither.
- a. odd
 - b. even
 - c. neither
 - d. both even and odd
- 15 Classify $f(x) = 3x^5 + 6x$ as even, odd, or neither.
- a. even
 - b. neither
 - c. odd
 - d. both even and odd
- 16 Classify $f(x) = -2x^2 + 7$ as even, odd, or neither.
- a. odd
 - b. both even and odd
 - c. neither
 - d. even
- 17 Classify $f(x) = 3x^3 + 5x$ as even, odd, or neither.
- a. even
 - b. odd
 - c. both even and odd
 - d. neither

18 Classify $f(x) = -2x^3 + 8$ as even, odd, or neither.

- a. neither
- b. odd
- c. both even and odd
- d. even

19 Classify $f(x) = 2x^2 + 4x$ as even, odd, or neither.

- a. both even and odd
- b. odd
- c. neither
- d. even

20 Classify $f(x) = -3x^3 + 4x$ as even, odd, or neither.

- a. even
- b. neither
- c. both even and odd
- d. odd

21 Classify $f(x) = -3x^3 + 4$ as even, odd, or neither.

- a. odd
- b. both even and odd
- c. neither
- d. even

22 Classify $f(x) = -x^4 + 8x^2$ as even, odd, or neither.

- a. odd
- b. both even and odd
- c. even
- d. neither

23 Classify $f(x) = -2x^4 + 8x^2$ as even, odd, or neither.

- a. odd
- b. neither
- c. even
- d. both even and odd

Multiple choice — medium

24 Find the domain of $f(x) = \sqrt{5x - 35}$.

- a. $x \geq 9$
- b. $x \geq 7$
- c. $x \geq 5$
- d. $x \leq 7$

25 Find the domain of $f(x) = \sqrt{x + 1}$.

- a. $x \geq 1$
- b. $x \leq -1$
- c. $x \geq -1$
- d. $x \geq -3$

26 Find the domain of $f(x) = \sqrt{x-1}$.

a. $x \geq -1$

b. $x \geq 1$

c. $x \leq 1$

d. $x \geq 3$

27 Find the domain of $f(x) = \sqrt{4x-24}$.

a. $x \geq 4$

b. $x \leq 6$

c. $x \geq 8$

d. $x \geq 6$

28 Find the domain of $f(x) = \sqrt{4x-20}$.

a. $x \geq 7$

b. $x \leq 5$

c. $x \geq 5$

d. $x \geq 3$

29 Find the domain of $f(x) = \sqrt{x+3}$.

a. $x \leq -3$

b. $x \geq -5$

c. $x \geq -3$

d. $x \geq -1$

30 Find the domain of $f(x) = \sqrt{-x + 5}$.

a. $x \geq 5$

b. $x \leq 5$

c. $x \leq 7$

d. $x \leq 3$

31 Find the domain of $f(x) = \sqrt{5x + 25}$.

a. $x \geq -5$

b. $x \geq -7$

c. $x \leq -5$

d. $x \geq -3$

32 Find the domain of $f(x) = \sqrt{4x - 28}$.

a. $x \geq 9$

b. $x \geq 7$

c. $x \leq 7$

d. $x \geq 5$

33 Find the domain of $f(x) = \sqrt{5x - 20}$.

a. $x \geq 6$

b. $x \leq 4$

c. $x \geq 2$

d. $x \geq 4$

34 Find the domain of $f(x) = \sqrt{5x}$.

- a. $x \geq 0$
- b. $x \geq 2$
- c. $x \leq 0$
- d. $x \geq -2$

35 Find the domain of $f(x) = \sqrt{-3x - 6}$.

- a. $x \leq -4$
- b. $x \leq 0$
- c. $x \leq -2$
- d. $x \geq -2$

36 Find the domain of $f(x) = \sqrt{-2x + 12}$.

- a. $x \leq 6$
- b. $x \leq 8$
- c. $x \leq 4$
- d. $x \geq 6$

37 Find the domain of $f(x) = \sqrt{5x + 40}$.

- a. $x \geq -10$
- b. $x \geq -6$
- c. $x \geq -8$
- d. $x \leq -8$

38 Find the domain of $f(x) = \sqrt{-3x - 27}$.

- a. $x \leq -11$
 - b. $x \geq -9$
 - c. $x \leq -7$
 - d. $x \leq -9$
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Multiple choice — hard

39 Let $p(x) = 2x + 1$ for $x < -2$, and $p(x) = 3x - 6$ for $x \geq -2$. Find $p(-6)$ and $p(-2)$.

- a. $-12, -11$
- b. $-11, -14$
- c. $-11, -12$
- d. $-9, -12$

40 Let $p(x) = 3x - 6$ for $x < 3$, and $p(x) = -2x + 3$ for $x \geq 3$. Find $p(-2)$ and $p(7)$.

- a. $-12, -11$
- b. $-12, -13$
- c. $-10, -11$
- d. $-11, -12$

41 Let $p(x) = 3x - 6$ for $x < 2$, and $p(x) = 2x - 4$ for $x \geq 2$. Find $p(1)$ and $p(6)$.

- a. $8, -3$
- b. $-3, 6$
- c. $-3, 8$
- d. $-1, 8$

- 42 Let $p(x) = x + 6$ for $x < 0$, and $p(x) = -5x + 6$ for $x \geq 0$. Find $p(-5)$ and $p(5)$.
- a. 1, -19
 - b. -19, 1
 - c. 3, -19
 - d. 1, -21
- 43 Let $p(x) = 4x - 7$ for $x < 5$, and $p(x) = -x + 1$ for $x \geq 5$. Find $p(3)$ and $p(5)$.
- a. 5, -6
 - b. 5, -4
 - c. -4, 5
 - d. 7, -4
- 44 Let $f(x) = x^2$ and $g(x) = f(x + 4) - 3 = (x + 4)^2 - 3$. Find $g(-5)$.
- a. -6
 - b. 4
 - c. 2
 - d. -2
- 45 Let $f(x) = x^2$ and $g(x) = 4f(x - 3) - 1 = 4(x - 3)^2 - 1$. Find $g(4)$.
- a. 7
 - b. -1
 - c. -3
 - d. 3

46 Let $f(x) = x^2$ and $g(x) = -2f(x - 4) - 6 = -2(x - 4)^2 - 6$. Find $g(3)$.

a. 8

b. -12

c. -8

d. -4

47 Let $f(x) = x^2$ and $g(x) = -f(x - 2) + 8 = -(x - 2)^2 + 8$. Find $g(-8)$.

a. -96

b. -92

c. -88

d. 92

48 Let $f(x) = x^2$ and $g(x) = 4f(x + 2) + 6 = 4(x + 2)^2 + 6$. Find $g(-8)$.

a. 150

b. 146

c. -150

d. 154

49 Find the domain of $g(x) = \frac{1}{x^2 - 64}$.

a. 8, -10

b. -8, 8

c. 10, -8

d. 8, -8

50 Find the domain of $g(x) = \frac{1}{x^2 - 144}$.

a. 14, -12

b. 12, -12

c. -12, 12

d. 12, -14

51 Find the domain of $g(x) = \frac{1}{x^2 - 16}$.

a. 4, -4

b. 6, -4

c. -4, 4

d. 4, -6

52 Find the domain of $g(x) = \frac{1}{x^2 - 49}$.

a. 9, -7

b. 7, -9

c. 7, -7

d. -7, 7

53 Find the domain of $g(x) = \frac{1}{x^2 - 9}$.

a. 5, -3

b. 3, -3

c. 3, -5

d. -3, 3